

# Building Legal RAG: what each component contributes

## 01 CONTEXT AND HYPOTHESIS

Retrieval-augmented generation (RAG) grounds language-model answers in external documents [1]. Legal QA needs correct answers and precise, complete source-page citations [2, 3]. PDFs may lack searchable text; evidence spans different text scales and may require exact legal terms. Retrieving more passages can also introduce irrelevant pages.

How can a legal RAG pipeline improve answer quality while keeping source-page citations both precise and complete?

|                                                                                                                                 |                                                                                                                                          |                                                                                                                                                          |
|---------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>H1 · Retrieval</b><br>Multiple text scales, hybrid search and reranking will improve answers and evidence retrieval over R0. | <b>H2 · Scan robustness</b><br>Recovering text from scanned documents will reduce the performance loss caused by the missing text layer. | <b>H3 · Citation selection</b><br>Selecting pages after answer generation will improve citation precision while retaining most of the relevant evidence. |
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## 02 METHODOLOGY

**Data:** 30 DIFC legal PDFs (590 pages), 100 questions: 70 structured-answer and 30 free-text. Reference source pages are available for 95 questions. In the mixed corpus, 10 PDFs (107 pages) become synthetic scans: page images at 300 DPI with the text layer removed.

**R0 · Baseline. R1–R6 each include all earlier additions.** PDF text → 300-token passages → semantic retrieval → GPT-5.4-mini → answer + retrieved pages as citations. Semantic search uses Qwen3-Embedding-0.6B.

| Document processing                                                                  |                                                                                                                              | Retrieval                                                                                        |                                                                                            | Answering                                                                                                        | Citations                                                                                                              |
|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>R1 · Read scans</b><br>Recover searchable text from page images.<br>OCR: PP-OCRv5 | <b>R2 · Multiple text scales</b><br>Represent evidence at different levels.<br>Pages, sections, clauses, microchunks, tables | <b>R3 · Find exact terms</b><br>Combine semantic and lexical search.<br>BM25 + rank fusion (RRF) | <b>R4 · Rank evidence</b><br>Prioritize the most relevant passages.<br>Qwen3-Reranker-0.6B | <b>R5 · Check answer type</b><br>Normalize and validate the expected type.<br>Type checks, confidence thresholds | <b>R6 · Refine citations</b><br>Select pages linked to supporting passages.<br>Generator's supporting-chunk references |

**Fixed:** GPT-5.4-mini / high, 24 retrieval candidates, ≤ 2,400 context tokens and the generation schema. R4–R6 reuse one generated response. R0–R6 run on the mixed corpus; R0/R1 also on originals.

**Evaluation:** Correct answers can still have poor citations. We score answers and citations separately.

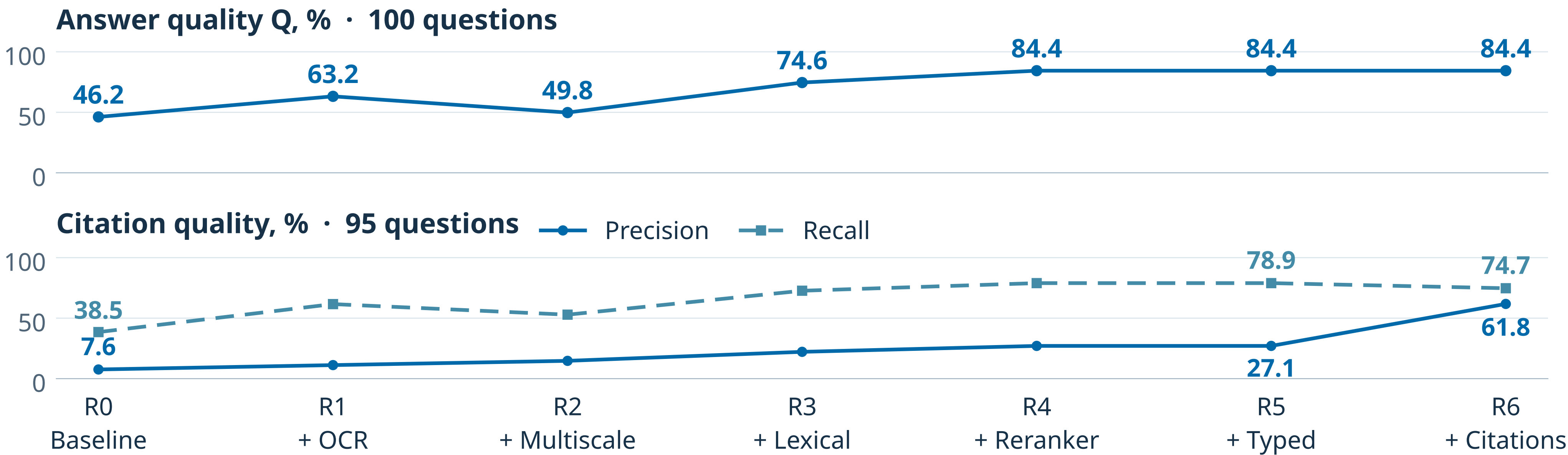
**Answer quality  $Q = 0.7 \times S + 0.3 \times F$ .** S / F: mean structured / free-text scores. Free text is LLM-scored.

**Citation precision:** share of cited pages that match annotated source pages. Are the citations relevant?

**Citation recall:** share of annotated source pages cited. Is the evidence complete? Both average 95 questions.

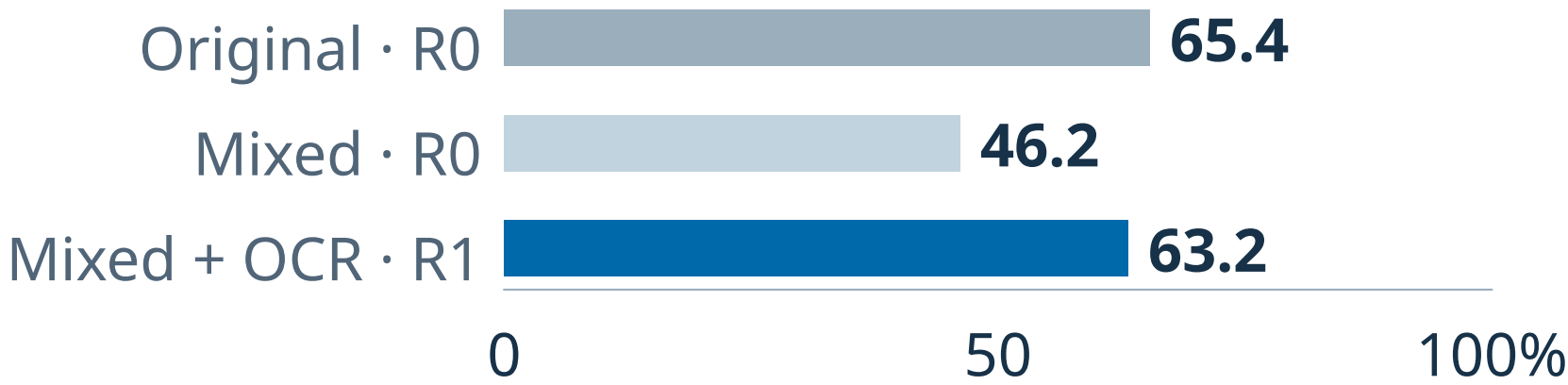
## 03 RESULTS

R0 to R6 on the mixed corpus



Q is an aggregate score, not the share of fully correct answers. R5 adds no score gain.

### OCR recovers performance on scans



Scanning 10 of 30 PDFs lowers Q from 65.4% to 46.2%. OCR restores most of this loss, reaching 63.2%.

### Hypothesis evaluation

**H1 · Partially supported.** Multiple scales alone reduce Q; lexical retrieval and reranking improve Q and recall.

**H2 · Supported.** OCR restores most of the loss caused by removing the text layer.

**H3 · Supported with a trade-off.** Precision rises from 27.1% to 61.8%; recall falls from 78.9% to 74.7% (–4.2 pp).

## Conclusion

OCR restores scans, hybrid retrieval and reranking improve answers, and citation selection trades recall for precision. Evaluate and optimize answer quality and citations separately.

**Limitations:** one DIFC development corpus, 100 questions and synthetic scans; sequential additions; one run per condition, no significance testing; LLM-scored free text.

**References:** [1] Lewis et al. RAG (2020). [2] Pipitone & Alami. LegalBench-RAG (2024). [3] Gao et al. ALCE (2023).

**Code, protocol & results:** [github.com/SteinVR/Trends-in-NLP-Poster](https://github.com/SteinVR/Trends-in-NLP-Poster)